



Inada Naoki

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How to speed up Python application startup time

#python

I hear [pipenv](#) 9.0.2 is released with major startup time improvement.

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Discover fresh viewpoints in this insightful post, supported by our vibrant DEV Community. **Every developer's experience matters**—add your thoughts and help us grow together. I tried it soon and I didn't feel it's fast. So I investigated it with Python 3.7's new feature.

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Startup time $\equiv$ import time

Okay

For example, execution time of `pipenv -h` is much larger than time to show help message.

Generally speaking, when application starts, there are some startup process like loading environment variables or config files.

In case of Python application, importing module takes most of startup time. For example, `pipenv --version` took about 800ms and `import pipenv` took 700ms.

```
$ time local/py37/bin/python3 -c 'import pipenv'
```

```
real    0m0.696s
user    0m0.648s
sys     0m0.048s
```

```
$ time local/py37/bin/pipenv --version
pipenv, version 9.0.3
```

```
real    0m0.812s
user    0m0.761s
sys     0m0.052s
```

Show import time for modules

Python 3.7 has new feature to show time for importing modules.

This feature is enabled with `-X importtime` option or `PYTHONPROFILEIMPORTTIME` environment variable.

For example, you can profile import time of pipenv by

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Investigating import time

Okay

At last of the output, you can see these lines:

```
import time: self [us] | cumulative | imported package
...
import time:          3246 |          578972 | pipenv.cli
import time:           507 |          579479 | pipenv
```

In last line, 579479 means `import pipenv` took 579 479us.

While importing `pipenv`, many other modules are imported. From above example, you can see `pipenv imports pipenv.cli`. Subimports are indented with 2 spaces.

See last line again. 507 means only 507us was took when running `pipenv` module. $579\,479 - 507 = 578\,972$ us is used for subimports.

Finding slow part

Let's find slow subtree from the output. I selected some lines.

```
import time: self [us] | cumulative | imported package
...
import time:          86500 |          179327 | pkg_resources
...
import time:           385 |          236655 | IPython
import time:           22 |          236677 | IPython.core
import time:           26 |          236703 | IPython.core.magic
import time:           978 |          237680 | dotenv.ipython
import time:          199 |          239032 | dotenv
```

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pkg_resources

A simple “thank you” can uplift the author and spark new discussions—leave yours below! As you can see, importing `pkg_resources` is slow. And surprisingly, `pkg_resources` is not indented; it's not subimport of `pipenv`.

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Okay

```
$ cat local/py37/bin/pipenv
#!/home/inada-n/local/py37/bin/python3.7
# EASY-INSTALL-ENTRY-SCRIPT: 'pipenv==9.0.3','console_scripts','pipe
```

```
__requires__ = 'pipenv==9.0.3'
import re
import sys
from pkg_resources import load_entry_point

if __name__ == '__main__':
    sys.argv[0] = re.sub(r'(-script\.pyw?|\\.exe)?$', '', sys.argv[0])
    sys.exit(
        load_entry_point('pipenv==9.0.3', 'console_scripts', 'pipenv
    )
```

Bad news: Importing `pkg_resources` is slow. It's known issue and it's difficult to fix without breaking backward compatibility.

Good news: you can avoid importing `pkg_resources`!

```
$ local/py37/bin/pip install wheel
$ local/py37/bin/pip install -U --force-reinstall pipenv
$ time local/py37/bin/pipenv --version
pipenv, version 9.0.3

real    0m0.704s
user    0m0.653s
sys     0m0.052s
```

When wheel is installed, pip builds wheel and install from it.

Installing from wheel (.whl) and from source package (.tar.gz) are different

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When installing from wheel, `pkg_resources` is not used in script:

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```
$ cat local/py37/bin/pipenv
#!/home/inada-n/local/py37/bin/python3.7
```

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```
# -*- coding: utf-8 -*-
import re
import sys
```

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```
if __name__ == '__main__':
```

Okay

```
sys.argv[0] = re.sub(r'(-script\.pyw?|\\.exe)?$', '', sys.argv[0])
sys.exit(cli())
```

IPython

See next part.

import time:	385	236655	IPython
import time:	22	236677	IPython.core
import time:	26	236703	IPython.core.magic
import time:	978	237680	dotenv.ipython
import time:	199	239032	dotenv

pipenv imports dotenv, dotenv imports dotenv.ipython, and it imports IPython.

That's why pipenv starts slow on my environment; I have IPython installed.

But why IPython is imported? I read dotenv source and I found it's for IPython's extension.

Surely, pipenv and many dotenv users don't use IPython extension.

I made [pull request](#) to dotenv which make importing IPython on demand.

And since pipenv has own copy of dotenv, I made [pull request](#) to pipenv which removes dotenv.ipython completely.

Conclusion

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I can reduce time for pipenv --version from 800ms to 500ms.
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```
$ time local/py37/bin/pipenv --version
pipenv, version 9.0.3
```

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```
real    0m0.503s
user    0m0.463s
sys     0m0.040s
```

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Import time profiling is very nice way to investigating and optimize application startup time.

Okay

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Okay

Dan Ryan •



Thanks for putting this article together and most importantly, thanks for contributing back to the project!

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Inada Naoki

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LOCATION

Japan

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EDUCATION

Osaka Prefecture University College of Technology

WORK

Tech Lead / OSS developer at KLab Inc.

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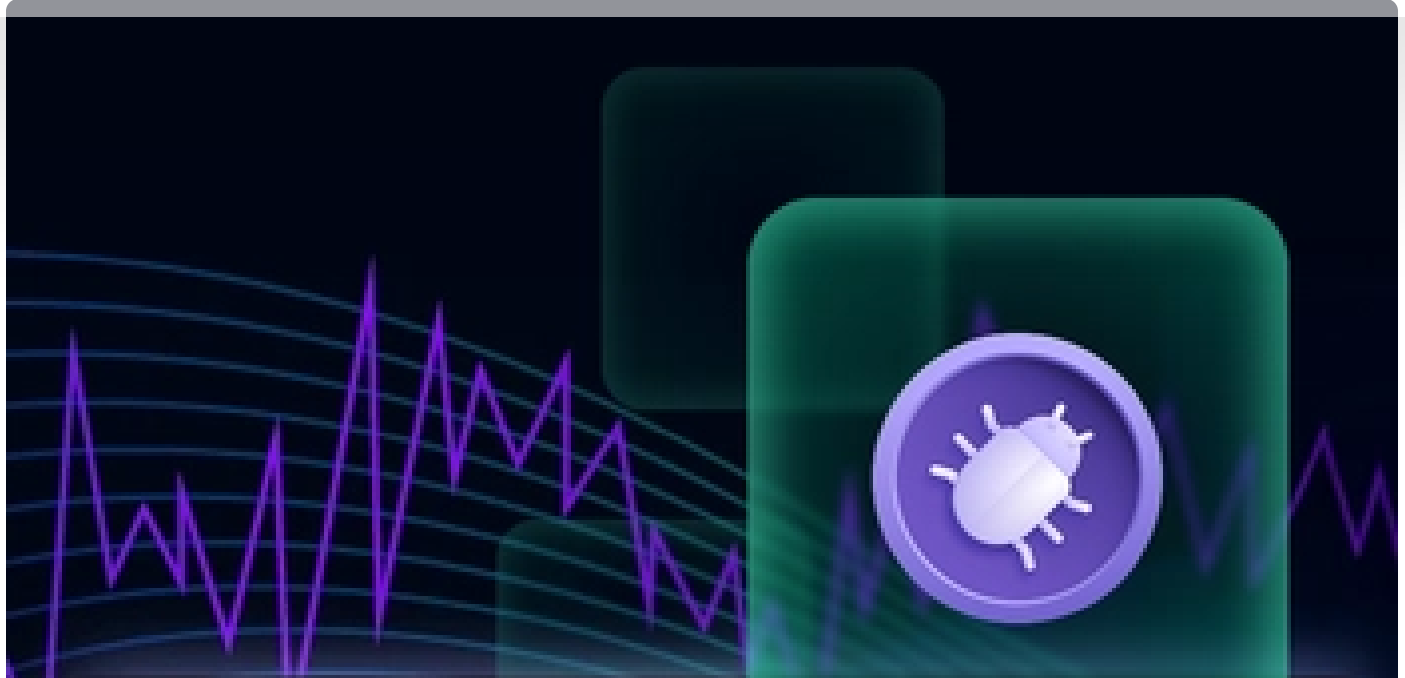
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